NEPA Project Analysis: Decarbonization Technology Environmental Reviews

National Environmental Policy Act Text Corpus (NEPATEC) 2.0 Analysis

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2026-08-21

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1 Project Overview

This document is a good reference for how to contextualize this project, the project goals and data, and to clearly understand the project deliverables.

1.1 Project context: National Environmental Policy Act (NEPA)

Federal environmental permitting—and, relatedly, the National Environmental Policy Act (NEPA)—has often been blamed as a core contributor to delays in infrastructure deployment. However, research to date on NEPA has been hindered by federal agency data management practices: NEPA documents are scattered across numerous agency databases, often in machine-unreadable formats, and typically lack basic metadata and other identifiers. As such, researchers have had to craft bespoke subsets of NEPA data from which to glean insights—a time-consuming task that has limited the information available about NEPA's effectiveness and paved the way for a national permitting reform conversation rife with anecdotes and cherry-picked information.

The newly released National Environmental Policy Act Text Corpus (NEPATEC) 2.0 dataset from the Pacific Northwest National Laboratory's (PNNL's) PermitAI project has the potential to add new, more comprehensive evidence into this conversation. PNNL has built and released a comprehensive dataset of past environmental reviews and permitting documents containing millions of pages and billions of words in machine-readable JSON format. The database contains more than 120,000 NEPA documents[1] from 60,000 projects prepared by more than 60 different agencies. Each document contains metadata for (as applicable):

- Lead agency
- Category
- Type of review
- Name of project
- Location
- Project sponsor
- Project sector
- Project type (a subset of project sector)
- Type of document
- Document title
- Agency or contractor responsible for preparing the document
- Categorical exclusion category
- Summary of the proposed action

> **Note**
>
> The National Environmental Policy Act Text Corpus (NEPATEC) 2.0 can be accessed here on huggingface.

[1] Including, for example, categorical exclusions, draft and final environmental assessments, draft and final environmental impact statements, records of decision, findings of no significant impact, and other supporting documentation.

1.2 Data structure

This shows the original data structure of the project in its original json format. The code in this repo manipulates this raw data.


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        "project_sector": {
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        "location": {
            "value": ""
        }
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        },
        "process_type": {
            "value": ""
        },
        "lead_agency": {
            "value": ""
        }
    },
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                    "document_ID": {
```

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            "section_or_volume_title": {
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            "file_provider": {
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            "page text": "PAGE 1 TEXT"
        },
        {
```

```
                    "page number": 2,
                    "page text": "PAGE 2 TEXT"
                }
            ]
        }
    ]
}
```


2 Project deliverables

For this project, we created tables, figures, and maps that answer the following questions across two phases of work.

2.1 Phase 1

1. **Data on number of decarbonization technology projects within the dataset**: number of projects broken down by technology (e.g., offshore and onshore wind, solar, geothermal, nuclear), lead agency, and location
2. **Data on programmatic and tiered reviews**: how many tiered reviews are there compared to total and are they completed faster
3. **Data on how many decarbonization technology projects have been categorically excluded vs. have required environmental assessments and environmental impact statements**
    - Broken out by number of projects, generation capacity, and change over time
4. **Data on geography/project location**: whether projects are multi-state or multi-agency
5. **Number of pages over time**, including pre- and post- Fiscal Responsibility Act of 2023 (FRA), which set page limit requirements
6. **Technology-specific inquiries**
    - Transmission lines: length of lines from project summary correlated with timelines, location, etc.
    - Geothermal: timelines of environmental reviews for different phases of the same project
    - Carbon and hydrogen pipelines (if available): length of pipelines from project summary correlated with timelines, location, etc.; compare to natural gas pipelines

2.2 Phase 2

Phase 2 was scoped on the basis of what Phase 1 established, and comprises the following deliverables:

1. **Reasons why NEPA was triggered** (e.g., federal land, federal funding) for different types of projects
2. **Determinations of significance across resource areas**; factors that contribute to a determination of "significant impact"
    - Starting with mitigated FONSIs
3. **Differences and similarities between NEPA reviews** for fossil fuel and decarbonization technology projects, as well as linear and non-linear projects—application of categorical exclusions, timelines, geography, etc.
4. **Timelines for categorical exclusions, environmental assessments, and environmental impact statements**, including segmentation by years (e.g., pre- and post-FRA [which set timelines for reviews], different CEQ NEPA regulations, agency, and type of project)
5. **Categorical exclusion spikes**: how the use of categorical exclusions responded to major federal funding laws, attributed through the statutes cited in the documents themselves
6. **Technical support for new regulatory categorical exclusion development**: identifying patterns in FONSIs

3 Project Structure

The directorates for each phase of this project are organized as follows:

Top level:

- `technical_reports/`: Cross-phase technical companion reports for Phase 1 and Phase 2
- `presentations/`: RevealJS slide decks for stakeholder meetings
- `app/`: Streamlit document explorer for browsing the underlying document text
- `literature/`: Reference materials and background documents
- `includes/`, `scripts/`: Site-wide HTML includes and Quarto render hooks
- `docs/`: The built Quarto website that publishes everything below

`phase1/` — the first analysis layer:

- `code/`: Python extraction scripts and per-deliverable R analysis

- `data/`: Raw and processed data files (parquet format) organized by review type (CE, EA, EIS)
- `reports/`: Quarto deliverable reports
- `output/`: Analysis outputs including tables, figures, and maps
- `notes/`: Status files and architecture notes
- `runbooks/`: Step-by-step pipeline instructions

`phase2/` — the extended analysis layer:

- `code/`: DuckDB-based extraction pipelines and per-deliverable scripts
- `data/`: Phase 2 outputs, written separately from Phase 1 data
- `reports/`: Quarto deliverable reports
- `output/`: Deliverable outputs, including the report-input tables
- `architecture/`: As-built technical documentation for each deliverable
- `factsheets/`: Client-facing one-page summaries for each deliverable
- `notes/`: Published methods notes and coverage pages
- `rag/`: Retrieval-augmented question answering over the document corpus
- `runbooks/`: Pipeline runbooks
- `training/`: Model training labels and locked sample identifiers

4 Definitions: What Qualifies as Decarbonization Technology?

4.1 Background

This section defines the universe of decarbonization technology projects analyzed across all deliverables of the NEPA Decarbonization Technology Analysis. It describes the classification criteria, exclusion rules, and refinements applied to identify decarbonization technology projects within the publicly released NEPATEC 2.0 database — a comprehensive record of federal environmental review activity under the National Environmental Policy Act (NEPA).

Understanding this classification framework is foundational: every project count, timeline, agency breakdown, and geographic pattern reported in subsequent deliverables flows from the decisions documented here.

4.2 The NEPA Universe

Figure 1 gives a sense of the total number of projects in the NEPATEC 2.0 database by NEPA review process type.

Only a portion of that review activity is relevant to this analysis. The corpus is therefore narrowed to projects involving decarbonization technology, using the project-type tags and refinements described in the sections below.

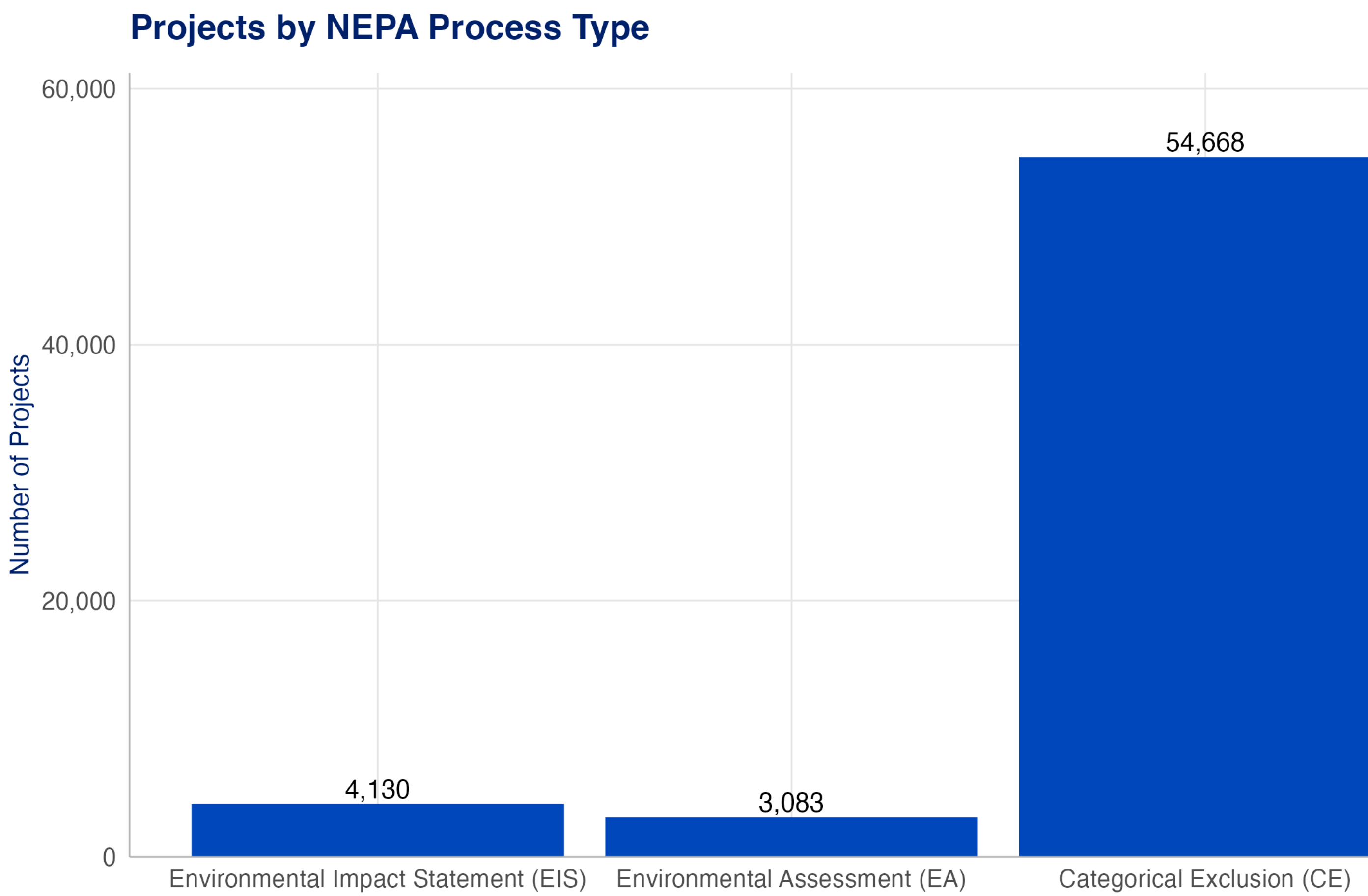

Figure 1: Distribution of all projects by review process in the NEPA database.

Figure 2 shows the decarbonization technology subset, totaling approximately **25,000 decarbonization technology projects** extracted from the NEPA database.

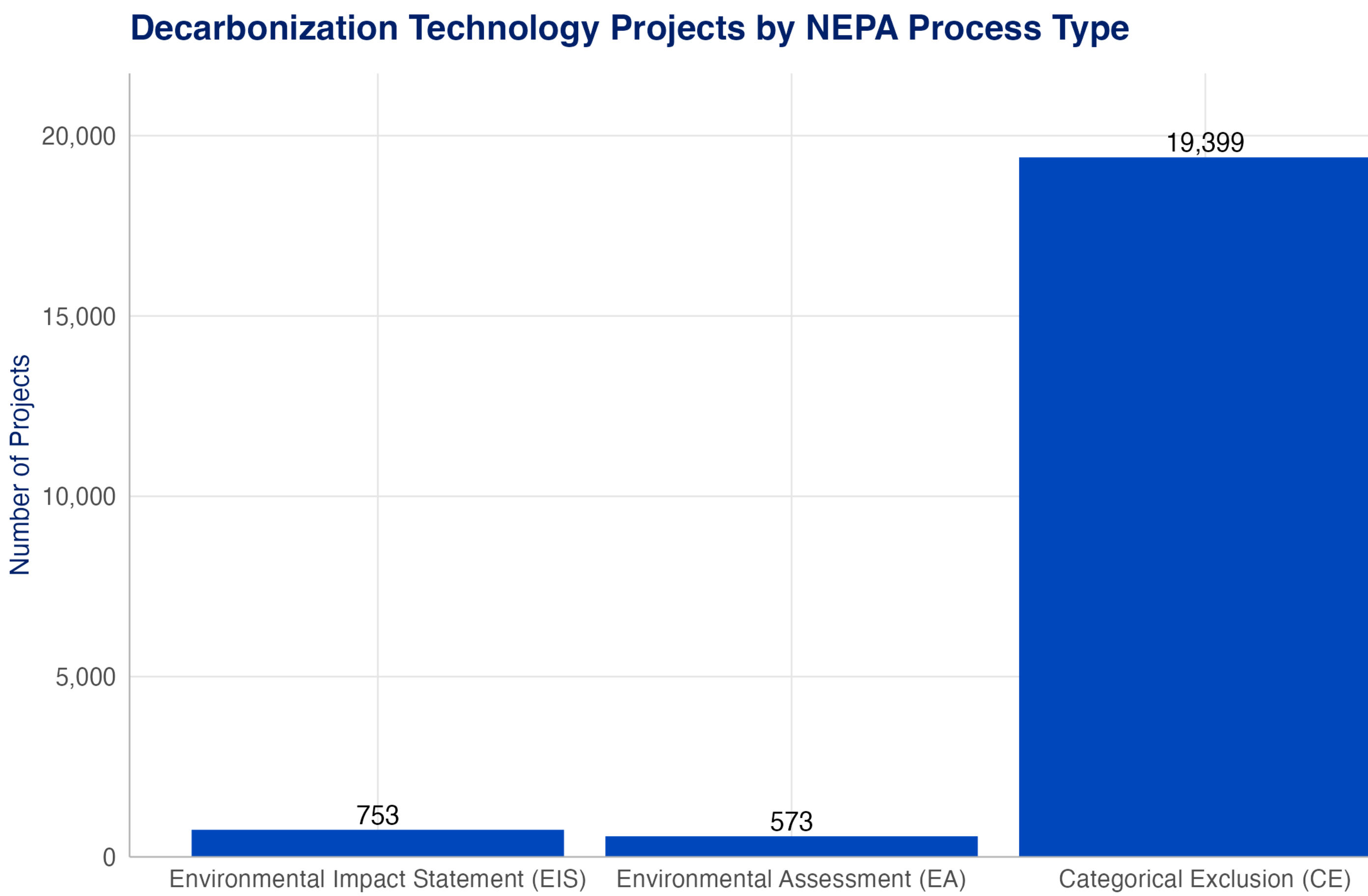

Figure 2: Distribution of decarbonization technology projects by review process in the NEPA database.

4.3 What Qualifies as Decarbonization Technology?

Table 1 enumerates the CATF-defined decarbonization technology and fossil fuel project type tags used to identify decarbonization technology vs fossil fuel energy projects in the NEPA database. The table also reports project counts for each tag or tag family. While projects often have multiple tags, a "decarbonization technology" project is identified as having at least one decarbonization technology tag *AND* no fossil fuel tags. As a result, the total count by technology type will always be greater than the total count by energy type (e.g., see Figure 3) because multi-tag technology projects appear in more than one technology category.

Table 1: Project type tags used to classify decarbonization vs. fossil energy technologies

Project Type Tag	Projects
Decarbonization Technology	
Utilities (electricity, gas, telecommunications)	10,157
Renewable Energy Production[1]	8,798
Electricity Transmission	7,697
Carbon Capture and Sequestration	1,327
Nuclear Technology	1,314
Conventional Energy Production - Other	935
Conventional Energy Production - Nuclear	301
Fossil Fuel	
Conventional Energy Production - Land-based Oil & Gas	8,664
Pipelines	5,324
Conventional Energy Production - Coal	732
Conventional Energy Production - Offshore Oil and Gas	222
Conventional Energy Production - Rural Energy	157

[1]Includes Biomass, Energy Storage, Geothermal, Hydrokinetic, Hydropower, Solar, Wind (Offshore & Onshore), and Other

Figure 3 shows the overall distribution of projects in the NEPATEC 2.0 database by energy type. Decarbonization technology projects comprise roughly 34% of all projects in the database, while fossil fuel projects represent just over 17%, and other (non-energy) projects make up 49% of the total. This breakdown helps contextualize the decarbonization technology subset analyzed in subsequent deliverables within the broader universe of federal environmental reviews.

About **10% of projects have only decarbonization technology tags**, but most (about 90%) have at least 1 decarbonization technology tag plus some other combination of tags.

4.3.1 Refining Decarbonization Technology

After reviewing a table of all co-occurring project types, three refinements were applied to improve the precision of the "decarbonization technology" classification:

- **Utilities + non-energy: 1,623 projects** tagged *ONLY* as Utilities *AND* with 1 or more of the following non-energy tags (e.g., broadband, waste management, land development) were excluded, since these likely reflect utility-adjacent infrastructure rather than energy generation.

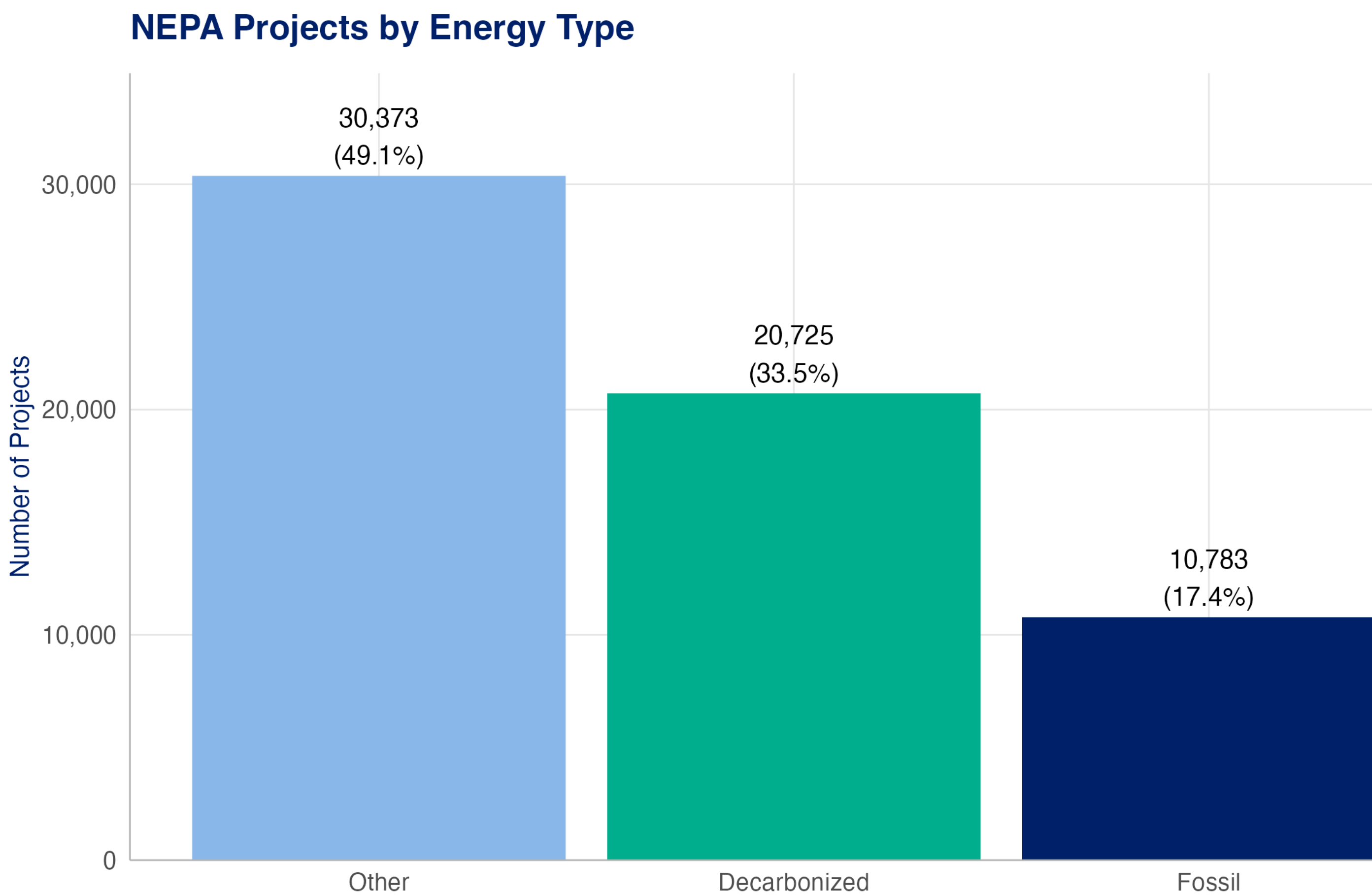

Figure 3: Distribution of all NEPA projects by energy type classification.

- **Military and Defense + Nuclear: 481 projects** tagged as "Nuclear Energy" and "Military and Defense" were excluded from the decarbonization technology category. A full table can be viewed here. The majority of these projects are led by the Department of Energy rather than the Department of Defense, suggesting they involve weapons-related nuclear activities rather than energy production.
- **Nuclear + Waste Management:** We started with approximately **4,000 Nuclear Waste projects** (those tagged with "Waste Management" AND ("Nuclear Technology" OR "Conventional Energy Production - Nuclear")), excluded all projects sponsored by DOE's National Nuclear Security Administration (NNSA), Office of Environmental Management (EM), and Office of Legacy Management (LM), along with their associated field offices. This reduced the dataset to approximately **1,588 projects**, and then CAFT staff identified only **34** projects that were relevant to this analysis. You can see a list of the 34 nuclear-tagged projects that remain in this analysis here.